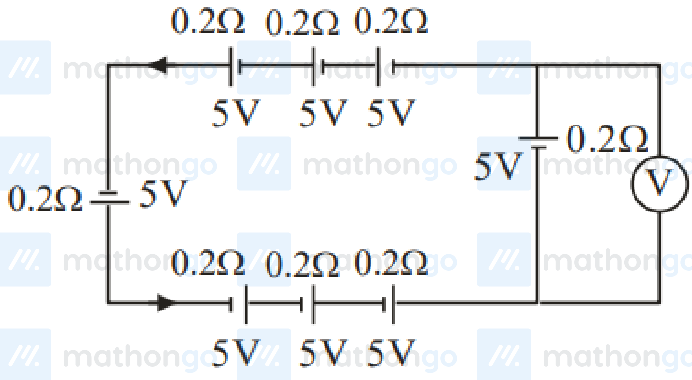
Q1 - 2024 (01 Feb Shift 1)

The reading in the ideal voltmeter (V) shown in the given circuit diagram is :



(1) 5 V

(2) 10 V

(3) 0 V

(4) 3 V

Q2 - 2024 (01 Feb Shift 1)

A galvanometer has a resistance of 50Ω and it allows maximum current of 5 mA. It can be converted into voltmeter to measure upto 100 V by connecting in series a resistor of resistance

(1) 5975Ω

(2) 20050Ω

(3) 19950Ω

(4) 19500Ω

Q3 - 2024 (01 Feb Shift 1)

The current in a conductor is expressed as $I=3 t^2+4 t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t=1$ s to t $=2$ s is_____ C.

Q4 - 2024 (01 Feb Shift 2)

In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G, the resistance of ammeter will be :

[We changed options. In official NTA paper no option was correct.]

(1) $\frac{G}{20}$

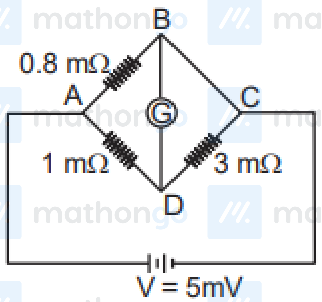
(2) $\frac{G}{199}$

(3) 199G

(4) 200G

Q5 - 2024 (01 Feb Shift 2)

To measure the temperature coefficient of resistivity α of a semiconductor, an electrical arrangement shown in the figure is prepared. The arm BC is made up of the semiconductor. The experiment is being conducted at $25^{\circ}\mathrm{C}$ and resistance of the semiconductor arm is $3\ \mathrm{m}\Omega$. Arm BC is cooled at a constant rate of $2^{\circ}\mathrm{C/s}$. If the galvanometer G shows no deflection after 10 s, then α is :



(1) $-2 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

(2) $-1.5 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

(3) $-1 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

(4) $-2.5 \times 10^{-2}{}^{\circ}\mathrm{C}^{-1}$

Q6 - 2024 (01 Feb Shift 2)

In a metre-bridge when a resistance in the left gap is 2Ω and unknown resistance in the right gap, the balance length is found to be 40 cm. On shunting the unknown resistance with 2Ω, the balance length changes by :

(1) 22.5 cm

(2) 20 cm

(3) 62.5 cm

(4) 65 cm

Q7 - 2024 (27 Jan Shift 1)

A wire of length 10 cm and radius $\sqrt{7} \times 10^{-4}$ m connected across the right gap of a meter bridge. When a resistance of 4.5Ω is connected on the left gap by using a resistance box, the balance length is found to be at 60 cm from the left end. If the resistivity of the wire is $\mathrm{R} \times 10^{-7} \Omega \mathrm{m}$, then value of R is :

(1) 63

(2) 70

(3) 66

(4) 35

Q8 - 2024 (27 Jan Shift 1)

A wire of resistance R and length L is cut into 5 equal parts. If these parts are joined parallely, then resultant resistance will be :

(1) $\frac{1}{25}\mathrm{R}$

(2) $\frac{1}{5}\mathrm{R}$

(3) 25R

(4) 5R

Q9 - 2024 (27 Jan Shift 2)

Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L, radius r. If X is the resistance of wire, then specific resistance is : $S_1 = X\left(\frac{\pi r^2}{L}\right)$. If the length of the wire gets doubled then the value of specific resistance will be :

(1) $\frac{S_1}{4}$

(2) $2\ \mathrm{S}_1$

(3) $\frac{S_1}{2}$

(4) S_1

Q10 - 2024 (27 Jan Shift 2)

A current of $200 \mu \mathrm{A}$ deflects the coil of a moving coil galvanometer through 60°. The current to cause deflection through $\frac{\pi}{10}$ radian is :

(1) $30 \mu \mathrm{A}$

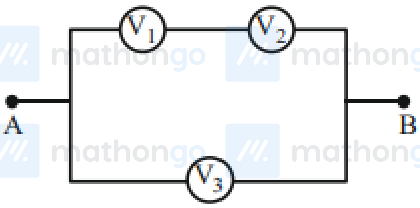
(2) $120 \mu \mathrm{A}$

(3) $60 \mu \mathrm{A}$

(4) $180 \mu \mathrm{A}$

Q11 - 2024 (27 Jan Shift 2)

Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1, V_2 and V_3. Choose the correct option.



(1) $V_1=V_2$

(2) $V_1 \neq V_3-V_2$

(3) $V_1+V_2>V_3$

(4) $V_1+V_2=V_3$

Q12 - 2024 (29 Jan Shift 1)

The electric current through a wire varies with time as $I=I_0+\beta$ t. where $I_0=20$ A and $\beta=3$ A/s. The amount of electric charge crossed through a section of the wire in 20 s is :

(1) 80C

(2) 1000C

(3) 800C

(4) 1600C

Q13 - 2024 (29 Jan Shift 1)

A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3 mA. For it to measure a current of 8 A, the value of the shunt should be:

(1) $3 \times 10^{-3}\Omega$

(2) $4.85 \times 10^{-3}\Omega$

(3) $3.75 \times 10^{-3}\Omega$

(4) $2.75 \times 10^{-3}\Omega$

Q14 - 2024 (29 Jan Shift 1)

The deflection in moving coil galvanometer falls from 25 divisions to 5 division when a shunt of 24Ω is applied. The resistance of galvanometer coil will be :

(1) 12Ω

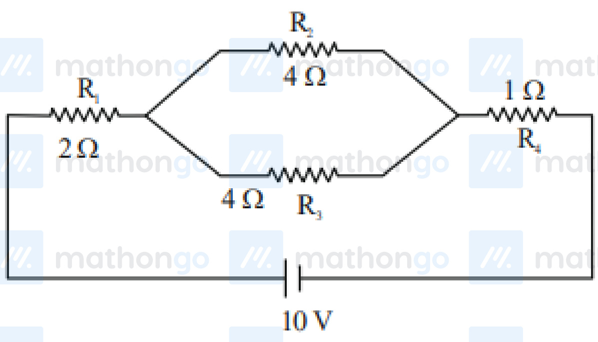
(2) 96Ω

(3) 48Ω

(4) 100Ω

Q15 - 2024 (29 Jan Shift 2)

In the given circuit, the current in resistance R_3 is :



(1) 1 A

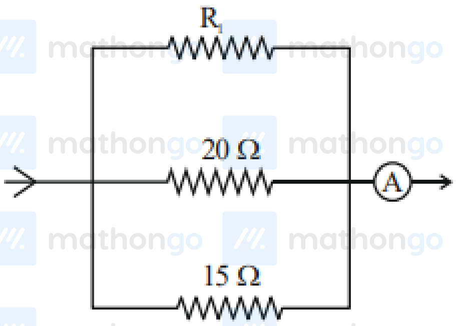
(2) 1.5 A

(3) 2 A

(4) 2.5 A

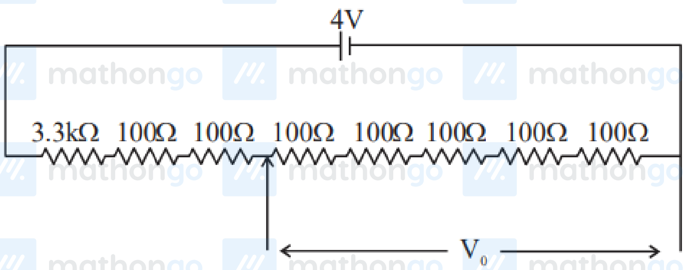
Q16 - 2024 (29 Jan Shift 2)

In the given circuit, the current flowing through the resistance 20Ω is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is ______ Ω.



Q17 - 2024 (30 Jan Shift 1)

A potential divider circuit is shown in figure. The output voltage V_0 is



(1) 4 V

(2) 2mV

(3) 0.5 V

(4) 12mV

Q18 - 2024 (30 Jan Shift 1)

An electric toaster has resistance of 60Ω at room temperature $(27^{\circ}\mathrm{C})$. The toaster is connected to a 220 V supply. If the current flowing through it reaches 2.75 A, the temperature attained by toaster is around : (if

$\alpha = 2 \times 10^{-4}/^{\circ}C)$

(1) $694^{\circ}C$

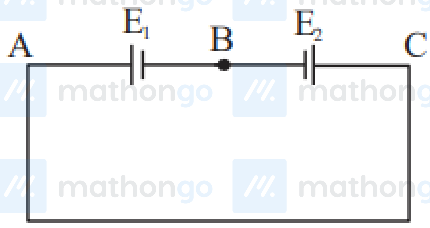
(2) $1235^{\circ}C$

(3) $1694^{\circ}C$

(4) $1667^{\circ}C$

Q19 - 2024 (30 Jan Shift 1)

Two cells are connected in opposition as shown. Cell E_1 is of 8 V emf and 2Ω internal resistance; the cell E_2 is of 2 V emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is:



Q20 - 2024 (30 Jan Shift 2)

When a potential difference V is applied across a wire of resistance R, it dissipates energy at a rate W. If the wire is cut into two halves and these halves are connected mutually parallel across the same supply, the same supply, the energy dissipation rate will become:

(1) 1/4 W

(2) 1/2 W

(3) 2 W

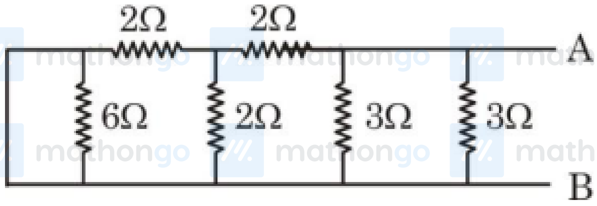
(4) 4 W

Q21 - 2024 (30 Jan Shift 2)

Two resistance of 100Ω and 200Ω are connected in series with a battery of 4 V and negligible internal resistance. A voltmeter is used to measure voltage across 100Ω resistance, which gives reading as 1 V. The resistance of voltmeter must be_____ Ω.

Q22 - 2024 (31 Jan Shift 1)

Equivalent resistance of the following network is_____ Ω.



Q23 - 2024 (31 Jan Shift 2)

The resistance per centimeter of a meter bridge wire is r, with $X\Omega$ resistance in left gap. Balancing length from left end is at 40 cm with 25Ω resistance in right gap. Now the wire is replaced by another wire of 2r resistance per centimeter. The new balancing length for same settings will be at

(1) 20 cm

(2) 10 cm

(3) 80 cm

(4) 40 cm

Q24 - 2024 (31 Jan Shift 2)

By what percentage will the illumination of the lamp decrease if the current drops by 20% ?

(1) 46%

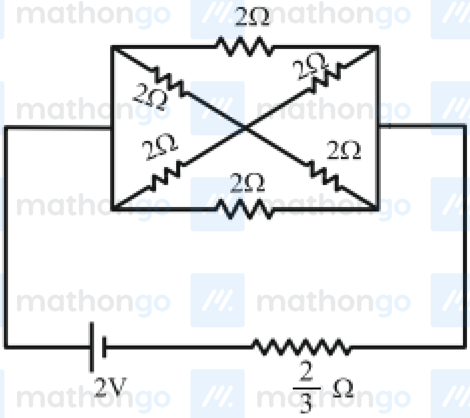
(2) 26%

(3) 36%

(4) 56%

Q25 - 2024 (31 Jan Shift 2)

In the following circuit, the battery has an emf of 2 V and an internal resistance of $\frac{2}{3}\Omega$. The power consumption in the entire circuit is_____ W.


2Ω
2Ω
2Ω
2Ω
2Ω
2Ω
2V
2/3 Ω

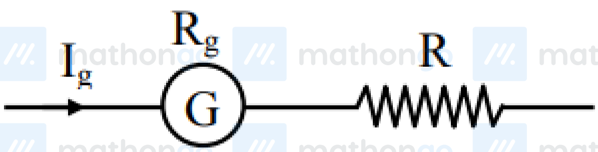
Answer Key

Q1 (3)	**Q2** (3)	**Q3** (22)	**Q4** (1)
Q5 (3)	**Q6** (1)	**Q7** (3)	**Q8** (1)
Q9 (4)	**Q10** (3)	**Q11** (4)	**Q12** (2)
Q13 (3)	**Q14** (2)	**Q15** (1)	**Q16** (30)
Q17 (3)	**Q18** (3)	**Q19** (6)	**Q20** (4)
Q21 (200)	**Q22** (1)	**Q23** (4)	**Q24** (3)
Q25 (3)			

Q1

$$i=\frac{E_{eq}}{r_{eq}}=\frac{8\times 5}{8\times 0.2}$$
$$I=25\text{ A}$$
$$V=E-ir$$
$$=5-0.2\times 25$$
$$=0$$

Q2

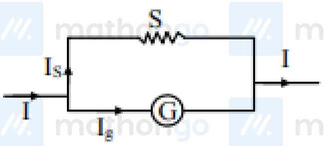


$$R=\frac{V}{I_g}-R_g=\frac{100}{5\times 10^{-3}}-50$$
$$=20000-50$$
$$=19950\Omega$$

Q3

$$q=\int_1^2 idt=\int_1^2\left(3t^2+4t^3\right)dt$$
$$q=\left(t^3+t^4\right)\Big|_1^2$$
$$q=22C$$

Q4



$$I_S S=I_g G$$
$$\frac{95}{100}IS=\frac{5I}{100}G$$
$$S=\frac{G}{19}$$

$R_A = \frac{SG}{S+G} = \frac{\frac{G^2}{19}}{\frac{20G}{19}}$

$R_A = \frac{G}{20}$

Q5

For no deflection $\frac{0.8}{1} = \frac{R}{3}$

$\Rightarrow R = 2.4\ m\Omega$

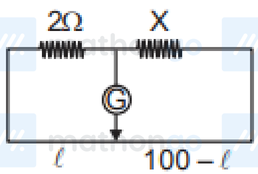
Temperature fall in 10 s $= 20^\circ C$

$\Delta R = R\alpha\Delta t$

$\alpha = \frac{\Delta R}{R\Delta t} = \frac{-0.6}{3 \times 20}$

$= -10^{-2} C^{-1}$

Q6



First case $\frac{2}{40} = \frac{X}{60} \Rightarrow X = 3\Omega$

In second case $X' = \frac{2\times3}{2+3} = 1.2\Omega$

$\frac{2}{\ell} = \frac{1.2}{100-\ell}$

$200 - 2\ell = 1.2\ell$

$\ell = \frac{200}{3.2} = 62.5$ cm

Balance length changes by 22.5 cm

Q7

For null point,

$\frac{4.5}{60} = \frac{R}{40}$

Also, $R = \frac{\rho\ell}{A} = \frac{\rho\ell}{\pi r^2}$

$4.5 \times 40 = \rho \times \frac{0.1}{\pi \times 7 \times 10^{-8}} \times 60$

$\rho = 66 \times 10^{-7}\Omega \times m$

Q8

Resistance of each part $= \frac{R}{5}$

Total resistance $= \frac{1}{5} \times \frac{R}{5} = \frac{R}{25}$

Q9

As specific resistance does not depends on dimension of wire so, it will not change.

Q10

$\mathrm{i} \propto \theta$ (angle of deflection)

$\therefore \frac{i_2}{i_1} = \frac{\theta_2}{\theta_1} \Rightarrow \frac{i_2}{200\mu\mathrm{A}} = \frac{\pi/10}{\pi/3} = \frac{3}{10}$

$\Rightarrow \mathrm{i}_2 = 60\mu\mathrm{A}$

Q11

From KVL,

$V_1 + V_2 - V_3 = \mathbf{0} \Rightarrow V_1 + V_2 = V_3$

Q12

Given that

Current $I = I_0 + \beta t$

$I_0 = 20$ A

$\beta = 3$ A/s

$\mathrm{I} = 20 + 3\mathrm{t}$

$\frac{\mathrm{dq}}{\mathrm{dt}} = 20 + 3\mathrm{t}$

$\int_0^{\mathrm{q}} \mathrm{dq} = \int_0^{20} (20 + 3\mathrm{t})\mathrm{dt}$

$\mathrm{q} = \int_0^{20} 20\mathrm{dt} + \int_0^{20} 3\mathrm{tdt}$

$\mathrm{q} = \left[20\mathrm{t} + \frac{3\mathrm{t}^2}{2}\right]_0^{20} = 1000\mathrm{C}$

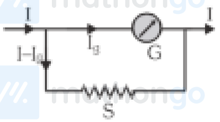
Q13

Given $G = 10\Omega$

$I_g = 3 \text{ mA}$

$I = 8 \text{ A}$

In case of conversion of galvanometer into ammeter.



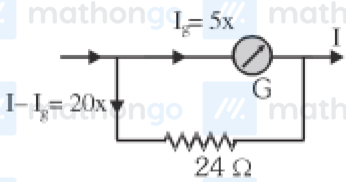
We have $I_g G = (I - I_g) S$

$$S = \frac{I_g G}{I - I_g}$$

$$S = \frac{(3 \times 10^{-3})\,10}{8 - 0.003} = 3.75 \times 10^{-3}\Omega$$

Q14

Let x = current/division



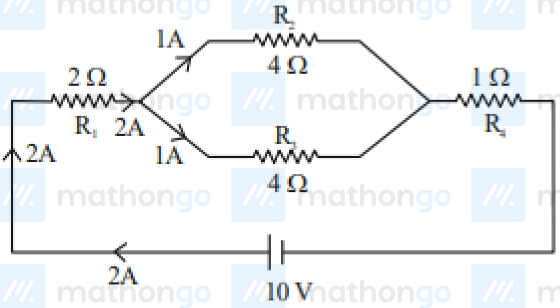
After applying shunt

Now $5x \times G = 20x \times 24$

$G = 4 \times 24$

$G = 96\Omega$

Q15



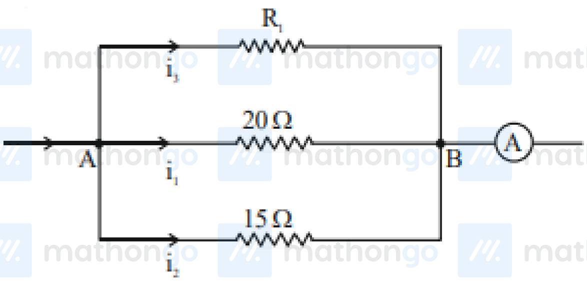
$R_{eq} = 2\Omega + 2\Omega + 1\Omega = 5\Omega$

$i = \frac{V}{R_{eq}} = \frac{10}{5} = 2\ A$

Current in resistance

$$\begin{aligned} R_3 &= 2 \times \left(\frac{4}{4+4}\right) \\ &= 2 \times \frac{4}{8} \\ &= 1\ A \end{aligned}$$

Q16



Given, $i_1 = 0.3\ A, i_1 + i_2 + i_3 = 0.9\ A$

So, $V_{AB} = i_1 \times 20\Omega = 20 \times 0.3\text{ V} = 6\text{ V}$

$\mathrm{i}_2 = \frac{6\text{ V}}{15\Omega} = \frac{2}{5}\text{ A}$

$\mathrm{i}_1 + \mathrm{i}_2 + \mathrm{i}_3 = \frac{9}{10}\text{ A}$

$\frac{3}{10} + \frac{2}{5} + \mathrm{i}_3 = \frac{9}{10}$

$\frac{7}{10} + \mathrm{i}_3 = \frac{9}{10}$

$\mathrm{i}_3 = 0.2\text{ A}$

So, $\mathrm{i}_3 \times \mathrm{R}_1 = 6\text{ V}$

$(0.2)\mathrm{R}_1 = 6$

$\mathrm{R}_1 = \frac{6}{0.2} = 30\Omega$

Q17

$R_{eq} = 4000\Omega$

$i = \frac{4}{4000} = \frac{1}{1000}\text{ A}$

$V_0 = i.R = \frac{1}{1000} \times 500 = 0.5\text{ V}$

Q18

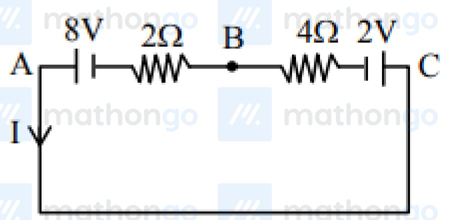
$\mathrm{R}_{\mathrm{T}=27} = 60\Omega, R_T = \frac{220}{2.75} = 80\Omega$

$\mathrm{R} = \mathrm{R}_0(1 + \alpha\Delta\mathrm{T})$

$80 = 60\left[1 + 2 \times 10^{-4}(\mathrm{T} - 27)\right]$

$\mathrm{T} \approx 1694^\circ\mathrm{C}$

Q19



$I = \frac{8-2}{2+4} = \frac{6}{6} = 1\text{ A}$

Applying Kirchhoff from C to B

$V_C - 2 - 4 \times 1 = V_B$
$V_C - V_B = 6 \text{ V}$
$= 6 \text{ V}$

Q20

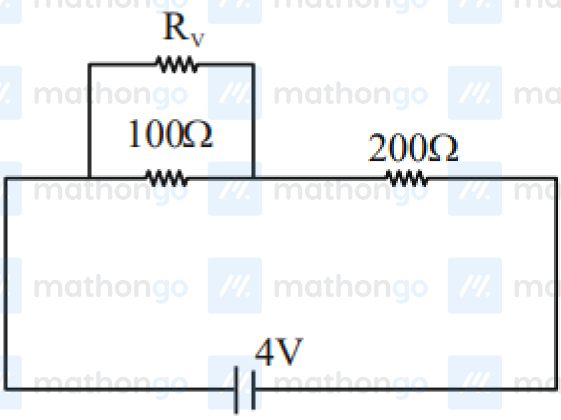
$\frac{v^2}{R} = W$ ...(i)

$\frac{v^2}{\frac{1}{2}\left(\frac{R}{2}\right)} = W'$ ...(ii)

From (i) \& (ii),

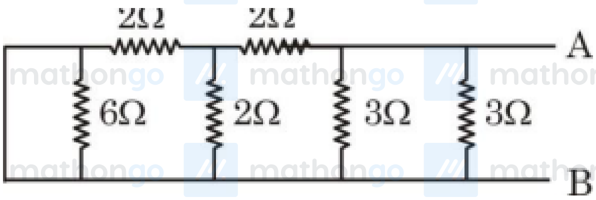
we get $W' = 4 \text{ W}$

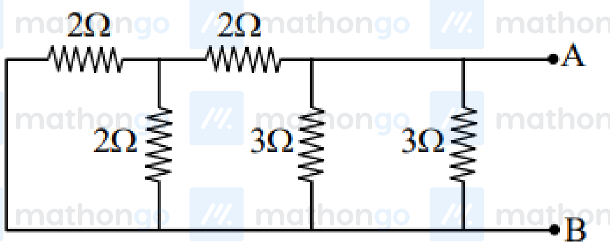
Q21



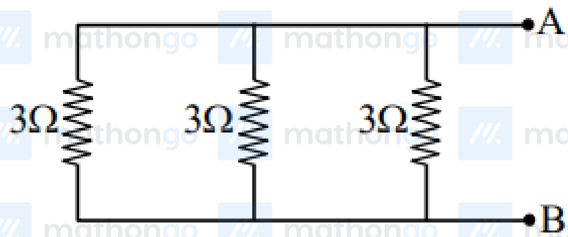
$$\frac{R_v 100}{R_v + 100} = \frac{200}{3}$$
$3R_v = 2R_v + 200$
$R_v = 200$

Q22



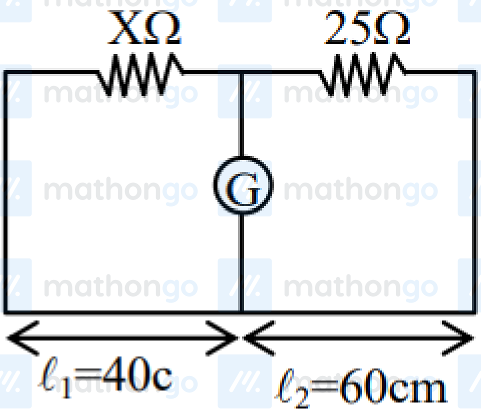


6Ω is short circuit



$R_{eq} = 3 \times \frac{1}{3} = 1 \Omega$

Q23



$\frac{25}{r\ell_1} = \frac{X}{r\ell_2}$ ...(i)

$\frac{25}{2r\ell_1'} = \frac{X}{2r\ell_2'}$

From (i) and (ii)

$\ell_2' = \ell_2 = 40$ cm

Q24

$P = i^2R$
$P_{\text{int}} = I_{\text{int}}^2 R$
$P_{\text{final}} = (0.8I_{\text{int}})^2R$
$\%$ change in power $=$
$\frac{P_{\text{final}} - P_{\text{int}}}{P_{\text{int}}} \times 100 = (0.64 - 1) \times 100 = -36\%$

Q25

$R_{eq} = \frac{4}{3}\Omega$

$\therefore P = \frac{V^2}{R_{eq}} = \frac{4}{4/3} = 3\text{ W}$